SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: **ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA01**

COURSE OUTCOME COVERED
CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)
Assessment Tool Weightage: 45 %

Total Marks: 50 **Time: 60 Mins**
No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I M.Varshitha Reddy,192525443 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	**Healthcare**	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	**Finance**	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	**Retail**	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	**Autonomous Vehicles**	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	**Robotics**	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	**Manufacturing**	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	**Title of AI Application**	Provide the name of the latest AI application selected and the corresponding AI domain.
2	**Domain Selection**	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	**Introduction and Background**	Describe the AI application and provide an overview of its purpose and importance.
4	**Problem Identification**	Explain the real-world problem or challenge addressed by this AI application.
5	**Need for AI Solution**	Analyze why AI is required and how it improves traditional approaches.
6	**AI Technologies Used**	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	**Working Mechanism**	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	**Data Sources and Processing**	Describe the type of data used and how the data is processed for intelligent decision-making.
9	**Real-World Implementation**	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	**Impact and Benefits**	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	**Challenges and Limitations**	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	**Future Scope and Emerging Trends**	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	**Conclusion**	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	**References**	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

1. Title of AI Application

Figure 02: An AI-Powered Humanoid Robot for Material Handling in BMW Manufacturing Plants

2. Domain Selection

Domain: Robotics

Specific Use Case: Parts transportation in BMW manufacturing plants and AI-based material handling

3. Introduction and Background

Figure 02 is an advanced AI-powered humanoid robot developed by Figure AI to perform human-like tasks in industrial and commercial environments. It combines artificial intelligence with robotics to understand its surroundings, make intelligent decisions, and perform complex physical tasks. The robot is designed with a human-like body structure, allowing it to work in environments built for people without requiring major infrastructure changes. Figure 02 integrates Computer Vision, Machine Learning, Deep Learning, Reinforcement Learning, and Natural Language Processing to interact with humans and perform tasks efficiently. One of its major applications is in BMW manufacturing plants, where it assists in material handling and logistics..

4. Problem Identification

Manufacturing industries face challenges such as labor shortages, repetitive manual work, workplace injuries, and increasing production demands. Workers are often required to transport heavy automobile parts between workstations, leading to physical fatigue and reduced productivity. Repetitive tasks can also increase the risk of human error and workplace accidents. As manufacturing becomes more complex, companies require intelligent systems capable of working continuously while maintaining high accuracy. Traditional automation systems are effective only for fixed and repetitive tasks, but they cannot easily adapt to dynamic factory environments. Figure 02 addresses these challenges by performing material handling tasks autonomously while collaborating safely with human workers.

5. Need for AI Solution

Traditional industrial robots operate based on predefined instructions and struggle to adapt when the environment changes. They cannot easily recognize new objects, understand human instructions, or make independent decisions. AI enables Figure 02 to perceive its surroundings, identify objects, learn from experience, and adjust its actions in real time. Computer Vision helps the robot recognize automobile parts, while Deep Learning improves object detection and decision-making accuracy. Reinforcement Learning allows the robot to optimize its movements over time. Natural Language Processing enables communication with human workers. By combining these AI technologies, Figure 02 performs tasks more efficiently, safely, and flexibly than traditional robotic systems.

6. AI Technologies Used

The Figure 02 humanoid robot uses several advanced Artificial Intelligence technologies:

- Artificial Intelligence (AI)
- Machine Learning (ML)

- Deep Learning (DL)
- Computer Vision
- Reinforcement Learning (RL)
- Natural Language Processing (NLP)
- Large Language Models (LLMs)
- Sensor Fusion
- Robotics and Motion Planning

7. Working Mechanism and System architecture

The working mechanism of Figure 02 consists of several stages:

1. Cameras and sensors capture images and environmental data.
2. Computer Vision identifies automobile parts, obstacles, and workstations.
3. Sensor data is processed and combined for accurate environmental understanding.
4. AI algorithms analyze the information and determine the required task.
5. Reinforcement Learning selects the safest and most efficient movement.
6. The robot uses its robotic hands to grasp the automobile component.
7. It walks autonomously to the target workstation while avoiding obstacles.
8. The component is placed accurately at the required location.
9. The robot receives the next instruction and repeats the process.

Components of the System Architecture

1. Input Data Collection Layer

The **Input Data Collection Layer** collects real-time information from the factory environment using multiple sensors. These sensors continuously gather data required for intelligent decision-making.

Components:

- RGB Cameras
- Depth Cameras
- Microphones
- Force Sensors
- IMU (Inertial Measurement Unit) Sensors
- Joint Position Sensors

Purpose:
The primary purpose of this layer is to capture visual images, detect objects, recognize voice commands, and monitor the robot's movement and posture within the manufacturing environment.

2. Data Processing Layer

The **Data Processing Layer** processes and prepares the collected sensor data before it is analyzed by Artificial Intelligence algorithms.

Components:

- Image Preprocessing
- Sensor Fusion
- Noise Removal
- Feature Extraction

Purpose:
This layer improves the quality, accuracy, and reliability of the collected data, enabling the AI system to make intelligent and accurate decisions.

3. Artificial Intelligence (AI) Layer

The **Artificial Intelligence Layer** serves as the brain of the Figure 02 humanoid robot. It analyzes the processed data, understands the surrounding environment, and makes intelligent decisions.

Computer Vision

Recognizes automobile parts, workers, workstations, shelves, and obstacles using images captured by the cameras.

Machine Learning

Learns from previous experiences and identifies patterns to improve task performance over time.

Deep Learning

Enhances object recognition, scene understanding, and environmental perception with high accuracy.

Reinforcement Learning

Learns the most effective movement strategies through continuous interaction with the environment and feedback.

Natural Language Processing (NLP)

Enables the robot to understand and respond to spoken instructions provided by human workers.

Large Language Model (LLM)

Supports advanced reasoning, natural communication, and task understanding during industrial operations.

4. Decision and Motion Planning Layer

After understanding the environment, the **Decision and Motion Planning Layer** determines the most efficient actions required to complete the assigned task.

Components:

- Task Planning
- Path Planning
- Object Grasp Planning
- Obstacle Avoidance

Purpose:
The purpose of this layer is to determine the safest, fastest, and most efficient sequence of actions while avoiding obstacles and ensuring smooth task execution.

5. Robot Control Layer

The **Robot Control Layer** converts AI-generated decisions into physical robot movements.

Components:

- Arm Controller
- Hand Controller
- Leg Controller
- Balance Controller

Purpose:
This layer ensures stable walking, accurate object grasping, precise placement of materials, and coordinated body movements during task execution.

6. Output Layer

The **Output Layer** represents the final execution of industrial tasks performed by the humanoid robot.

Outputs:

- Picks automobile parts
- Carries components between workstations
- Places materials at the correct workstation
- Interacts with workers through voice communication
- Receives the next task and continues operation

Purpose:
The Output Layer completes the assigned material handling tasks efficiently while supporting continuous industrial production and collaboration with human

7. Data Sources and Processing

Figure 02 relies on multiple data sources, including RGB cameras, depth cameras, force sensors, motion sensors, microphones, and joint position sensors. These sensors continuously collect information about the surrounding environment, object positions, human activities, and robot movements. The collected data is preprocessed to remove noise and improve quality. Computer Vision algorithms detect and classify objects, while sensor fusion combines information from different sensors to create a complete understanding of the environment. AI models analyze the processed data to make intelligent decisions regarding navigation, object manipulation, and task execution. The robot continuously updates its knowledge using real-time feedback to improve performance.

8. Real-World Implementation

Figure 02 has been implemented in collaboration with **BMW Manufacturing**, where it performs material handling and logistics tasks within automotive production facilities. The robot transports automobile components between different workstations, reducing repetitive manual labor. Beyond manufacturing, Figure AI aims to deploy Figure 02 in warehouse operations, logistics centers, healthcare support, retail inventory management, and household assistance. These real-world implementations demonstrate the robot's ability to operate safely in environments designed for humans while increasing productivity and operational efficiency.

9. Impact and Benefits

The implementation of Figure 02 provides several benefits. It increases production efficiency by performing repetitive tasks continuously without fatigue. The robot reduces workplace injuries by handling physically demanding activities. Its high precision minimizes errors during material transportation, improving manufacturing quality. Figure 02 supports human workers by taking over repetitive tasks, allowing employees to focus on more complex responsibilities. The robot can operate around the clock, increasing productivity while reducing long-term operational costs. Its adaptability also enables deployment across multiple industries beyond automotive manufacturing.

10. Challenges and Limitations

Despite its advantages, Figure 02 faces several challenges. The robot is expensive to develop, deploy, and maintain. Battery life limits continuous operation, requiring periodic recharging. AI systems must ensure safe interaction with humans to prevent accidents. Privacy and cybersecurity are important because the robot collects and processes large amounts of sensor data. Training AI models requires extensive datasets and computational resources. Additionally, unexpected environmental conditions and highly complex tasks remain difficult for humanoid robots, limiting their deployment in certain situations.

11. Future Scope and Emerging Trends

Future developments of Figure 02 include improved reasoning capabilities through advanced AI models, better battery technology for longer operation, and enhanced dexterity for performing delicate tasks. Researchers are working to expand its applications into healthcare, elderly assistance, disaster response, agriculture, construction, and household automation. Future humanoid robots are expected to collaborate more naturally with humans, understand complex instructions, and adapt to changing environments with minimal supervision. Continued advancements in AI, robotics, and sensor technology will further improve the capabilities and adoption of humanoid robots across industries.

12. Conclusion

Figure 02 represents a significant advancement in Artificial Intelligence and robotics by combining intelligent perception, autonomous decision-making, and human-like physical capabilities. Its deployment in BMW manufacturing plants demonstrates how AI-powered humanoid robots can improve productivity, safety, and operational efficiency. Although challenges such as cost, battery limitations, and ethical concerns remain, ongoing research and technological improvements are expected to overcome these issues. Figure 02 highlights the growing role of AI-powered humanoid robots in transforming manufacturing and many other industries, making it an important milestone in the future of intelligent automation.

13. References

1. Figure AI. (2024). *Figure 02: The Next Generation AI Humanoid Robot*.
2. BMW Group. (2024). *BMW Group Partners with Figure AI to Deploy Humanoid Robots in Manufacturing*.
3. OpenAI. (2024). *OpenAI and Figure AI Collaborate to Develop Intelligent Humanoid Robots*.
4. NVIDIA. (2024). *Isaac Platform for AI-Powered Humanoid Robots*.

5. IEEE Xplore Digital Library. (2024). *Humanoid Robotics and Artificial Intelligence in Smart Manufacturing*.
6. SpringerLink. (2024). *Artificial Intelligence and Humanoid Robots in Industrial Automation*.
7. ScienceDirect. (2024). *AI-Based Humanoid Robots for Industrial Applications*.
8. ACM Digital Library. (2024). *Recent Advances in Humanoid Robotics and Machine Learning*.
9. Nature Machine Intelligence. (2024). *Recent Advances in Artificial Intelligence for Robotics*.
10. Google Scholar. (2024). *Research Papers on Figure AI, Humanoid Robotics, and Artificial Intelligence*.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.